

Order-Zero Super Symbols Need Not Be Bounded on Any Schatten Class Other Than $\mathfrak{L}^2$

Automated Math Discovery – lane 9

July 19, 2026

Status and source question

Status. Candidate counterexample, likely valid pending human review.

This packet concerns the p -Schatten boundedness question in G. Lee and M. Lein, *A proof of $\mathfrak{L}^2$ -boundedness for magnetic pseudodifferential super operators via matrix representations with respect to Parseval frames*, arXiv:2405.19964, published in *Monatshefte für Mathematik* **206** (2025), 597–627 [1].

On page 3 the source asks for conditions under which a magnetic pseudodifferential super operator is bounded on the trace class and, more generally, on $\mathfrak{L}^p$ for $1 \leq p \leq \infty$:

quantization the operator will act from the left or the right.

That poses the question we wish to answer here: for generic functions L belonging to some Hörmander class, under what conditions on L does the associated magnetic pseudo-differential *super* operator define a bounded operator

$$\mathrm{Op}^A(L) : \mathfrak{L}^1(\mathcal{B}(L^2(\mathbb{R}^d))) \longrightarrow \mathfrak{L}^1(\mathcal{B}(L^2(\mathbb{R}^d))) \quad (1.1)$$

between trace class operators? More generally, when is

$$\mathrm{Op}^A(L) \in \mathcal{B}(\mathfrak{L}^p(\mathcal{B}(L^2(\mathbb{R}^d)))) \quad (1.2)$$

a bounded operator between p -Schatten classes where $1 \leq p < \infty$? Or we could pose the same question for the space of bounded operators $\mathfrak{L}^\infty(\mathcal{B}(L^2(\mathbb{R}^d))) = \mathcal{B}(L^2(\mathbb{R}^d))$ to itself, which corresponds to $p = \infty$.

As the notation suggests, these operator spaces are particular cases of non-commutative

Theorem 5.3 proves that every order-zero symbol in $S_{\rho,0}^0$ gives a bounded operator on $\mathfrak{L}^2$. Remark 5.4 (pages 29–30) singles out the hoped endpoint extensions under the same hypotheses, which would yield all other exponents by interpolation:

Remark 5.4 If we could prove that

$$\text{Op}^A(F) \in \mathcal{B}(\mathcal{B}(L^2(\mathbb{R}^d))) = \mathcal{B}(\mathfrak{L}^\infty(\mathcal{B}(L^2(\mathbb{R}^d))))$$

was bounded on $\mathcal{B}(L^2(\mathbb{R}^d))$ under the *same* conditions as in Theorem 5.3, the non-commutative version of the Riesz-Thorin Interpolation Theorem (cf. e. g. Theorems 2.9 and 2.10 as well as Remark 1 on p. 23 in [Sim05] or [GK69, Theorem 13.1]) would imply that

$$\text{Op}^A(F) : \mathfrak{L}^p(\mathcal{B}(L^2(\mathbb{R}^d))) \longrightarrow \mathfrak{L}^p(\mathcal{B}(L^2(\mathbb{R}^d)))$$

defined a bounded operator for any $2 \leq p \leq \infty$.

Likewise, a proof of boundedness on the trace-class operators $\mathfrak{L}^1(\mathcal{B}(L^2(\mathbb{R}^d)))$ would imply boundedness for any $1 \leq p \leq 2$.

The result below gives a sharp negative answer to that same-hypotheses extension. It does not attempt to classify the additional symbol conditions that do imply $\mathfrak{L}^p$ boundedness.

Idea of the proof

When the magnetic field is zero and a super symbol depends only on the two position variables, its super Weyl quantization multiplies operator kernels:

$$K(x, y) \longmapsto F(x, y)K(x, y).$$

Thus a smooth function of the difference $x - y$ produces a continuous Toeplitz Schur multiplier. We construct a sign sequence $m : \mathbb{Z} \rightarrow \{-1, 1\}$ whose Fourier multiplier is unbounded on every $L^p(\mathbb{T})$ except $L^2(\mathbb{T})$, and smoothly interpolate the values $m(n)$ by a function $\phi \in C^\infty(\mathbb{R})$ with every derivative bounded. The symbol

$$F(x_L, \xi_L, x_R, \xi_R) = \phi(x_L - x_R)$$

then belongs to every order-zero class $S_{\rho, 0}^0$.

Small, disjoint translates of one test function embed matrices into operators on $L^2(\mathbb{R})$. On this embedded copy, multiplication by $\phi(x - y)$ is exactly the Toeplitz Schur multiplier $(a_{ij}) \mapsto (m(i - j)a_{ij})$. Fourier–Schur transference now transfers the bad $L^p(\mathbb{T})$ multiplier behavior to every Schatten class. The one exceptional exponent is $p = 2$, where any bounded entrywise multiplier is automatically bounded.

Construction of one multiplier bad at every $p \neq 2$

We normalize Haar measure on $\mathbb{T}$ and write

$$D_N(t) = \sum_{k=1}^N e^{ikt}.$$

Lemma 1. *There is a real sequence $m : \mathbb{Z} \rightarrow \{-1, 1\}$ such that the Fourier multiplier*

$$T_m \left(\sum_k \widehat{f}(k) e^{ikt} \right) = \sum_k m(k) \widehat{f}(k) e^{ikt}$$

does not extend boundedly to $L^p(\mathbb{T})$ for any $p \in [1, \infty]$, $p \neq 2$.

Proof. Fix $q > 2$. Khintchine's inequality, applied pointwise in t and then integrated, shows that for every N there are signs $\varepsilon_1, \dots, \varepsilon_N$ such that

$$\left\| \sum_{k=1}^N \varepsilon_k e^{ikt} \right\|_{L^q(\mathbb{T})} \leq C_q N^{1/2}. \quad (1)$$

On the other hand, $|D_N(t)| \geq cN$ when $|t| \leq c/N$, and hence

$$\|D_N\|_{L^q(\mathbb{T})} \geq c_q N^{1-1/q}. \quad (2)$$

Enumerate the rational numbers in $(2, \infty)$, repeating every rational infinitely many times. For the j th term q_j , choose a fresh finite interval $I_j \subset \mathbb{Z}$, with length N_j tending to infinity sufficiently fast, and choose signs on I_j satisfying (1) for q_j . Define m to be those signs on the pairwise disjoint intervals I_j , and put $m(k) = 1$ elsewhere.

For a block I_j define

$$f_j(t) = \sum_{k \in I_j} m(k) e^{ikt}.$$

Up to multiplication by a character, (1) gives $\|f_j\|_{q_j} \leq C_{q_j} N_j^{1/2}$, whereas $m(k)^2 = 1$ and (2) give

$$\|T_m f_j\|_{q_j} \geq c_{q_j} N_j^{1-1/q_j}.$$

The norm ratio is at least a constant times $N_j^{1/2-1/q_j}$. The lengths can be chosen so that, along the infinitely many blocks assigned to each rational $q > 2$, this ratio tends to infinity. Thus T_m is unbounded on every rational $L^q(\mathbb{T})$ with $q > 2$.

Since $|m| = 1$, T_m is an isometry on L^2 . If it were bounded on some L^p with $p > 2$, interpolation with L^2 would make it bounded on every L^q with $2 < q < p$, contradicting a rational exponent in this interval. For $1 < p < 2$, boundedness would imply boundedness of the adjoint multiplier on $L^{p'}$; because m is real, the adjoint has the same symbol, contradicting the case $p' > 2$. Boundedness on L^1 or L^∞ , together with duality or interpolation with L^2 , likewise implies boundedness at some finite exponent different from 2. Therefore no such bounded extension exists. $\square$

The counterexample

Choose $\eta \in C_c^\infty((-1/4, 1/4))$, real-valued, with $0 \leq \eta \leq 1$ and $\eta = 1$ on $[-1/8, 1/8]$, and let

$$\phi(t) = \sum_{n \in \mathbb{Z}} m(n) \eta(t - n). \quad (3)$$

The supports in this sum are disjoint. Consequently ϕ is real-valued, bounded, smooth, and every derivative of ϕ is bounded.

Theorem 1. *Let $d = 1$, $B = 0$, and $A = 0$. The real super symbol*

$$F(x_L, \xi_L, x_R, \xi_R) = \phi(x_L - x_R) \quad (4)$$

belongs to $S_{\rho,0}^0(\mathbb{R}^4)$ and to the double class $S_{\rho,0}^{0,0}(\mathbb{R}^4)$ for every $0 \leq \rho \leq 1$. Its non-magnetic super Weyl quantization $\text{Op}^0(F)$ is bounded on $\mathfrak{L}^2(L^2(\mathbb{R}))$, but it has no bounded extension on $\mathfrak{L}^p(L^2(\mathbb{R}))$ for any $p \in [1, \infty]$, $p \neq 2$.

Proof. All position derivatives of F are bounded because all derivatives of ϕ are bounded, while every derivative in ξ_L or ξ_R vanishes. This is precisely the required order-zero estimate for both stated Hörmander classes, for every $\rho \in [0, 1]$.

For a symbol independent of the two momentum variables, the non-magnetic super Weyl quantization is joint position functional calculus. Hence, on smooth finite-rank operators X with integral kernel K_X ,

$$K_{\text{Op}^0(F)X}(x, y) = \phi(x - y)K_X(x, y). \quad (5)$$

This also follows first for tensor symbols from the identity

$$F(x_L, x_R) = f_L(x_L)f_R(x_R) \implies \text{Op}^0(F)X = f_L(Q)Xf_R(Q),$$

and then from the distributional super Weyl formula.

Choose a real unit vector $h \in C_c^\infty((-1/32, 1/32))$ and set $h_j(x) = h(x - j)$ for $j \in \mathbb{Z}$. These vectors have pairwise disjoint supports, so the map $U : \ell^2(\mathbb{Z}) \rightarrow L^2(\mathbb{R})$, $Ue_j = h_j$, is an isometry. If $a = (a_{ij})$ is a finite matrix, then UaU^* has kernel

$$K_a(x, y) = \sum_{i,j} a_{ij} h_i(x) \overline{h_j(y)}.$$

For $(x, y) \in \text{supp}(h_i) \times \text{supp}(h_j)$ we have $|(x - y) - (i - j)| < 1/16$. The plateau and disjoint-support properties of η in (3) therefore imply

$$\phi(x - y) = m(i - j).$$

Combining this identity with (5) gives the exact intertwining relation

$$\text{Op}^0(F)(UaU^*) = U(m(i - j)a_{ij})_{i,j \in \mathbb{Z}} U^*. \quad (6)$$

For finite p , $a \mapsto UaU^*$ preserves the Schatten p -norm. If $\text{Op}^0(F)$ were bounded on $\mathfrak{L}^p(L^2(\mathbb{R}))$, relation (6) would give a uniformly bounded Toeplitz Schur multiplier

$$S_m : (a_{ij}) \mapsto (m(i - j)a_{ij})$$

on $\mathfrak{L}^p(\ell^2(\mathbb{Z}))$. The Fourier-to-Schur transference inequality of Neuwirth and Ricard [2, Theorem 2.1] says, for the amenable group $\mathbb{Z}$, that the $L^p(\mathbb{T})$ Fourier multiplier norm of m is no larger than this Toeplitz Schur multiplier norm. Lemma 1 therefore rules out boundedness for every finite $p \neq 2$.

At $p = \infty$, boundedness together with the automatic $\mathfrak{L}^2$ bound would, by interpolation of Schatten classes, give boundedness on every $\mathfrak{L}^q$, $2 < q < \infty$, which has just been ruled out. Finally, $\text{Op}^0(F)$ is bounded on $\mathfrak{L}^2$ directly from (5): under the Hilbert–Schmidt kernel identification it is multiplication by the bounded function $\phi(x - y)$. $\square$

Verification, failed route, and scope

The first attempted positive route decomposed the source paper’s rapidly off-diagonal four-index frame matrix into shifts. For fixed shifts, however, the remaining coefficient array is an arbitrary Schur multiplier; a bounded array need not be bounded on $\mathfrak{L}^p$ when $p \neq 2$. This obstruction led to the construction above. An intermediate endpoint-only construction used the triangular projection. The final sign-block construction upgrades that route to one symbol failing simultaneously at every exponent $p \neq 2$.

The finite script `code/check_block_multiplier.py` samples the block mechanism. With $q = 4$, 128 trials, and a 16384-point quadrature grid, the observed Dirichlet-to-random-sign norm ratio increased from 1.70 at $N = 16$ to 3.14 at $N = 256$. This is only a sanity check; the proof is the Khintchine and Dirichlet-kernel estimate above.

The theorem disproves a universal $S_{\rho,0}^0 \Rightarrow \mathfrak{L}^p$ boundedness statement for every $p \neq 2$, already in the zero-field case. It does not exclude positive results for smaller symbol classes. Indeed, Cornean and Thorn [3] prove later sufficient conditions involving integrable weights (Proposition 5.2) and a completed projective tensor-product symbol class (Proposition 5.5). Those stronger subclasses do not contain the general order-zero conclusion refuted here.

Novelty check and human-review recommendation

The four lightweight run indexes were searched for arXiv:2405.19964, its title, “magnetic pseudodifferential super operator”, “Schatten”, “trace class”, “Schur multiplier”, and close variants; no duplicate was found. Bounded web searches used the exact source title and combinations of “order zero”, “Schatten p ”, “Schur multiplier”, and the authors’ names. They found the 2025 published source and the March 2026 follow-up arXiv:2603.26455, but no paper stating this counterexample or the sharp failure for all $p \neq 2$.

The recommended human review is to check two convention-sensitive points: (i) equation (5) under Lee–Lein’s right-action convention, and (ii) the application of Neuwirth–Ricard Theorem 2.1 to the full Toeplitz set for $\Gamma = \mathbb{Z}$. The remainder is elementary harmonic analysis and an isometric finite-matrix compression.

References

- [1] G. Lee and M. Lein, *A proof of $\mathfrak{L}^2$ -boundedness for magnetic pseudodifferential super operators via matrix representations with respect to Parseval frames*, Monatshefte für Mathematik **206** (2025), 597–627; arXiv:2405.19964.
- [2] S. Neuwirth and È. Ricard, *Transfer of Fourier multipliers into Schur multipliers and sumsets in a discrete group*, Canadian Journal of Mathematics **63** (2011), 1161–1187; arXiv:1001.5332.
- [3] H. D. Cornean and M. H. Thorn, *Magnetic Weyl super calculus: Schatten-class properties, commutator criterion, and complete positivity*, arXiv:2603.26455, 2026.